Thiago Ferreira dos Santos

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Last update: October 2026

Education

Yale University (US) – PhD candidate in Astrophysics — Aug. 2023 – Present

Universidade de São Paulo (Brazil) – MSc in Astronomy — Aug. 2021 – Jul. 2023
Thesis: Unveiling Planetary Systems in Solar Twins with High-Precision Radial Velocity

Universidade Federal de Santa Catarina (Brazil) – BSc in Physics — Mar. 2016 – Apr. 2021

Refereed Publications

Metrics: Full List and Refereed | Google Scholar

First-Author Publications

7. **Ferreira, T.** & Galarza, J.Y., et al., AJ, 172, 211 (2026)
   Giant Planets and Eccentric Orbits Are Common Around Galactic Thick Disc Stars
6. **Ferreira, T.** & Rice, M., AJ, 172, 111 (2026)
   On the Eccentricity Distribution and Tidal Evolution of Transiting Brown Dwarfs
5. **Ferreira, T.**, Bellinger, E.P., Farag, E., Lindsay, C.J., ApJ, 1001, 41 (2026)
   Red-Giant Asteroseismology of Low-Mass Population III Stars
4. **Ferreira, T.**, Bellinger, E.P., Farag, E., Lindsay, C.J., ApJ, 1000, 147 (2026)
   Evolution of Low-Mass Population III Stars: Convection, Mass Loss, Nucleosynthesis, and Neutrinos
3. **Ferreira, T.**, Rice, M., et al., AJ, 168, 145 (2024)
   SOLES XI. The Aligned Orbit of TOI-2533 b, a Transiting Brown Dwarf Orbiting an F8-type Star
2. **Ferreira, T.**, Saito, R.K., Minniti, D., et al., MNRAS, 527, 10737 (2024)
   A Benchmark White Dwarf–Ultracool Dwarf Wide Field Binary
1. **Ferreira, T.**, Saito, R.K., Minniti, D., et al., MNRAS, 486, 1220 (2019)
   The Asymptotic Evolution of the Stellar Merger V1309 Sco: a Blue Straggler in the Making?

Co-Author Publications

8. Winnick, I., et al. (incl. **Ferreira, T.**), accepted for publication in AJ (2025; arXiv:2508.16513)
   Extreme Lithium Depletion in Solar Twins: Challenging Non-Standard Mixing Models
7. Galarza, J.Y., Lorenzo-Oliveira, D., **Ferreira, T.**, et al., ApJ, 983, 70 (2025)
   HIP 8522: A Puzzling Young Solar Twin with the Lowest Detected Lithium
6. Fermiano, V., et al. (incl. **Ferreira, T.**), A&A, 690, L7 (2024)
   The Young Exoplanetary System TOI-4562: Confirming The Presence of a Third Body in the System
5. Galarza, J.Y., **Ferreira, T.**, Lorenzo-Oliveira, D., et al., AJ, 168, 91 (2024)
   TOI-1173 A b: The First Inflated Super-Neptune in a Wide Binary System
4. Galarza, J.Y., Reggiani, H., **Ferreira, T.**, et al., ApJ, 974, 122 (2024)
   Detailed Abundances of the Planet-Hosting TOI-1173 A/B: Possible Evidence of Planet Engulfment in a Very Wide Binary
3. Saito, R.K., et al. (incl. **Ferreira, T.**), A&A, 689, A148 (2024)
   The VISTA Variables in the Vía Láctea eXtended (VVVX) ESO Public Survey: Completion of the Observations and Legacy

2. Botan, E., et al. (incl. **Ferreira, T.**), MNRAS, 504, 654 (2021)
   Unveiling Short-Period Binaries in the Inner VVV Bulge

1. Herpich, F.R., et al. (incl. **Ferreira, T.**), A&A, 647, A169 (2021)
   VVV Survey Near-Infrared Colour Catalogue of Known Variable Stars

Non-Referred Publications and Conference Proceedings

10. **Ferreira, T.** et al., TASC9/KASC16 (2025);
    On the Structure and Oscillations of the First Stars

9. **Ferreira, T.** & Meléndez, J., BASBr, 34, 264 (2023);
   HIP 102152b: A low-mass planet candidate around an old solar twin

8. **Ferreira, T.** & Meléndez, J., The 21st Cambridge Workshop on Cool Stars (2021);
   HIP 102152b: A low-mass planet candidate around an old solar twin

7. **Ferreira, T.**, Saito, R.K., Minniti, D., ATel, 14626, 1 (2021);
   VVV near-IR observations of the X-ray transient Swift J174038.1-273712

6. **Ferreira, T.**, Saito, R.K., Minniti, D., ATel, 14625, 1 (2021);
   VVV near-IR observations of V6585 Sgr progenitor

5. **Ferreira, T.** & Saito, R.K., Planets 2020, id. 11 (2020);
   Exoplanetary transits in the Galactic bulge

4. Masetti, N., **Ferreira, T. S.**, et al., ATel, 11992, 1 (2018);
   The possible VVV near-infrared counterpart of IGR J17503-2636

3. **Ferreira, T.**, et al. RNAAS, 2, 39 (2018);
   VVV Survey Orbital Period Confirmation for the Cataclysmic Variable IGR J17014-4306

2. Kammers, R., **Ferreira, T.**, et al., ATel, 11365, 1 (2018);
   The likely progenitor of Nova ASASSN-18ds

1. **Ferreira, T.** & Saito, R.K., ATel, 10780, 1 (2017);
   The likely progenitor of Nova ASASSN-16ra

Telescope Observing Experience

ESO VLT / ESPRESSO – 1 night, Co-I 2026B (FTC 117A)

Radial Velocity Mass Confirmation of the Most Extreme Super-Puff around a Metal-Poor Thick-Disk Star with ESPRESSO

Gemini North/South 8.1-m / GHOST & MAROON-X – 1 night, Co-I 2026B

Confirming Galactic Chemical Imprints on Rocky Planet Interiors: A GHOST and MAROON-X Survey of Thick-Disk Exoplanet Hosts

Keck I 10-m / HIRES – 2 nights, PI 2026A

Uncovering Early Supernova Signatures in CEMP Halo Giant Stars

Subaru 8.2-m / HDS – 1.5 nights, PI 2026A

Uncovering Early Supernova Signatures in CEMP Giant Stars

Gemini South 8.1-m / GHOST – 1.5 nights, PI 2026A

Uncovering Early Supernova Signatures in CEMP Giant Stars

Magellan Clay 6.5-m / MIKE & PFS – 2 nights, Co-I 2026A

Exploring the Origin of Exoplanets in the Milky Way's Thick-Disk

Gemini North/South 8.1-m / GHOST & MAROON-X – 2 nights, Co-I 2026A

Exploring the Origin of Exoplanets in the Milky Way's Thick-Disk

Gemini South 8.1-m / GHOST – 0.5 nights, DDT 2025B

Confirming the First Exoplanet Around a Milky Way Halo Star

Magellan Clay 6.5-m / WINERED – 1 night, PI 2025A

Investigating Atmospheric Outflows in a Neptunian-desert Planet Using He I λ10 833 Å

Magellan Clay 6.5-m / PFS – 2 nights, Co-I 2025A

Confirming Thick Disc Membership for Known Exoplanets and TESS objects of interest

Magellan Clay 6.5-m / MIKE – 12 nights, Co-I 2024B, 2025A

Exploring the star–planet connection with solar-type hosts
Correlated Abundances of Exoplanets and Their Host Stars: Constraints from Polluted White Dwarfs in Stellar Binaries
Confirming Thick Disc Membership for Known Exoplanets and TESS objects of interest

ESO 3.6-m / HARPS – 6 nights, PI 2023A (P111)

How rare are analogues of the Solar System?

Seminars

Asteroseismology of Low-Mass Population III Stars – TOS group, HITS/Heidelberg (Germany) Jul 2025

Conference Talks & Posters

10th TESS/17th Kepler Asteroseismic Science Consortium Workshop, Aarhus, Denmark Jul 2026
Red-Giant Asteroseismology of Low-Mass Population III Stars (talk).
Robust Mass, Radius, and Age Measurement of a Nearly Pristine Star (poster; Grunblatt et al.).

Cool Stars 23, Tokyo, Japan Jun 2026
Robust Mass, Radius, and Age Measurement of a Nearly Pristine Star (poster; Grunblatt et al.).
A Homogeneous Census of Thick Disc Planetary Systems (poster).

9th TESS/16th Kepler Asteroseismic Science Consortium Workshop, Vienna, Austria Jul 2025
On the Structure and Oscillations of the First Stars (poster).

exoLaTam 22, Santiago, Chile Dec 2022
HIP 102152b: A Low-Mass Planet Candidate Around an Old Solar Twin (talk).

XLV Annual Meeting of the Brazilian Astronomical Society, Virtual Oct 2022
HIP 102152b: A Low-Mass Planet Candidate Around an Old Solar Twin (talk).

Cool Stars 21, Virtual Jul 2022
HIP 102152b: A Low-Mass Planet Candidate Around an Old Solar Twin (poster).

Exoplanets III, Virtual Jul 2021
A Box-Fitting Implementation for the Transit Method Detection of Exoplanets (poster).

Sagan Exoplanet Summer Workshop, Virtual Jul 2021
Time-Series Analysis of Near-IR Variabilities on VVV M Dwarf Stars (poster).

ALMA Planets2020 Workshop, Santiago Mar 2020
Exoplanetary Transits in the Galactic Bulge (poster).

EWASS 2019, Lyon, France Jun 2019
Infrared Counterparts of Gaia Alerts (poster; Ivanov et al.).

VVV & VVV-X Science Workshop, Florianópolis, Brazil Apr 2018
On the Photometric Study of the Cataclysmic Variable IGR J17014-4306 (poster).

Seminário de Iniciação Científica, UFSC, Virtual 2018, 2019, 2020, 2021

Stellar Variability Towards the Central Region of the Milky Way with the VVV/VVV-X Surveys (talk).

PPGFSC/UFSC Meeting, Florianópolis, Brazil — 2018, 2020
Stellar Variability Towards the Central Region of the Milky Way with the VVV/VVV-X Surveys (poster).

Fellowships, Awards & Grants

Yale Conference Travel Fund - Newton Endowment	Jul 2025, Jul 2026
GSAS Yale Graduate Fellowship	2023 – Present
Boris Garfinkel Prize Fellowship in Astronomy	2023
CAPES M.Sc. Research Fellowship, IAG/USP, Brazil	2021 – 2023
exoLaTam 2022 Travel Award	Dec 2022
Teaching Improvement Program, IAG/USP	2022
ESO/ALMA Planets2020 Workshop Travel Award	Mar 2020
CNPq Undergraduate Research Fellowship, UFSC, Brazil	2017 – 2021

Workshops & Advanced Schools

YCIU Primordial Black Holes Workshop, US	Mar 2026
Yale AstroCodEx Hack Day, New Haven	Aug 2025
MESA Summer School, Leuven, Belgium	Jul 2025
Yale Tinsley Workshop, New Haven	Oct 2024
New York Area Exoplanets Meeting, New York	May 2024
ESO Star–Planet Connection Workshop, Virtual	Oct 2021
XIII CBPF School, Virtual	Aug 2021
ZTF Summer School on Variable Stars, Virtual	Aug 2021
Sagan Exoplanet Summer Workshop, Virtual	Jul 2021

Service and Teaching

Referee – ApJ — 2025

Teaching Assistant – Yale University — Aug 2024 – May 2026
- ASTR 110 (Planets and Stars); Inst. Michael Faison
- ASTR 220 (Galaxies and Cosmology); Inst. Robert Zinn
- ASTR 556 (Astrostatistics and Data Mining); Inst. Earl Bellinger
- ASTR 310 (Galactic and Extragalactic Astronomy); Inst. Jeffrey Kenney

Teaching Assistant – IAG/USP — 2022
- AGA 0502 (Planets and Planetary Systems); Inst. Jorge Luis Melendez Moreno

Outreach

First-Year Astronomy Buddy (FAB) Mentorship Program Mentor — Aug 2026 – Present

Academic References

Earl P. Bellinger (PhD advisor) – Yale University — earl.bellinger@yale.edu
Jorge L. Meléndez (MSc advisor) – Universidade de São Paulo — jorge.melendez@iag.usp.br
Roberto K. Saito (BSc advisor) – Universidade Federal de Santa Catarina — robsaito@gmail.com
Dante Minniti – Universidad Andres Bello | Vatican Observatory — vvvdante@gmail.com